RESERVA HUARAVITO LTEMS

Primary Research Line 2026–2030

Phenothermic Synergy Hypothesis:

The Role of *Hauya elegans* as a Facilitator of Trophic Pulses within the Myrtaceae–Lauraceae Complex under Climatic Refuge Conditions

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ABSTRACT

This research proposal outlines a five-year investigation (2026–2030) into the mechanistic relationship between forest structural complexity, microclimatic buffering, and phenological stability in a premontane tropical forest fragment. We hypothesize that the exceptional thermal gradient documented at Huaravito Reserve (≈10°C differential between degraded and intact forest sectors) is causally linked to canopy architecture dominated by ***Hauya elegans*** (Onagraceae), and that this buffering effect facilitates enhanced fruiting constancy and seed viability in co-occurring Myrtaceae and Lauraceae taxa. The proposed framework integrates field phenology, distributed sensor networks, avifaunal documentation, and remote sensing validation to test a four-stage positive feedback model: structural complexity → thermal buffering → trophic magnetism → genomic enrichment. This research directly addresses SNAPP-MORE objectives by identifying low-cost, high-signal monitoring indicators that predict long-term restoration trajectories in tropical montane systems. We propose that microclimatic stability, measured via interannual temperature variance (σ ~0.2°C at Huaravito), represents a functional metric for climatic refugium status under regional warming scenarios.

Keywords: biogenic cooling, *Hauya elegans*, phenological stability, Myrtaceae, Lauraceae, microclimatic buffering, climatic refugium, tropical forest restoration, LTEMS

I. CONCEPTUAL FRAMEWORK

1.1 Theoretical Foundation

Tropical forest fragments embedded in agricultural matrices experience elevated thermal stress due to reduced canopy cover, altered wind regimes, and diminished evapotranspirative flux (Hardwick et al. 2015; de Frenne et al. 2019). These edge effects compromise reproductive phenology, seed viability, and recruitment dynamics, particularly for shade-tolerant, large-seeded taxa (Laurance et al. 2002). However, structurally intact forest nuclei with high basal area and multi-layered canopy architecture may maintain microclimatic homeostasis, creating "thermal islands" that buffer against regional climatic variability (Chen et al. 1999).

The **Phenothermic Synergy Hypothesis** proposes that such buffering is not merely a passive consequence of canopy closure but represents an active ecological feedback loop. Specifically, we hypothesize that:

- **Structural canopy dominants** (e.g., *Hauya elegans*) generate localized aerodynamic roughness, reducing wind velocity and enhancing moisture retention at understory levels.
- **Thermal stabilization** reduces metabolic stress on reproductive tissues, prolonging flowering and fruiting windows and increasing seed dormancy viability.
- **Phenological stability** attracts avian and mammalian dispersers, increasing heterospecific seed rain and genetic connectivity with external populations.
- **Enhanced recruitment** reinforces structural complexity, completing a positive feedback loop that sustains genomic diversity and ecosystem function under external climatic perturbation.

This framework positions **microclimatic buffering as a mechanistic driver of restoration success**, with direct implications for monitoring and adaptive management strategies.

1.2 Study System: Huaravito Reserve LTEMS

The Huaravito Reserve Sacred Natural Site (5,700 m² total; 4,000 m² mature forest core) is located in Quebrada Honda, Piedades Sur, San Ramón, Alajuela Province, Costa Rica at 1,250 m.a.s.l., classified as premontane humid forest (Holdridge). The site consists of:

- **Primary forest core** ("Toyopán Sacred Forest"): 4,000 m² mature canopy with high structural density of *Hauya elegans* (estimated 85–100 mature individuals; basal area 21–34 m²/ha for this species alone).
- **Regeneration zone**: 1,700 m² former pasture under passive restoration since 2021, exhibiting natural recruitment of baseline forest species.
- **Degraded reference area**: Former clay extraction zone (tajo) to the north-east, serving as thermal and ecological baseline for comparative monitoring.

Preliminary field measurements (February 2026) documented a **10.2°C temperature differential** between the degraded terrace (27.5°C) and the forest core interior at "The Shrine" (17.3°C) over approximately 150 meters' horizontal distance under midday clear-sky conditions. This magnitude exceeds typical tropical forest buffering ranges (3–8°C; de Frenne et al. 2019) and warrants systematic investigation.

II. CORE RESEARCH QUESTION

Does the thermal buffering associated with *Hauya elegans*-dominated canopy structure (observed differential ≈10°C between degraded and intact sectors) significantly enhance fruiting stability, seed viability, and germination success of key Myrtaceae and Lauraceae species at Huaravito Reserve LTEMS?

2.1 Focal Taxa

We target three species complexes representing critical trophic and successional guilds within the premontane Neotropical forest:

A. Myrtaceae Complex:

- *Myrcia splendens* (Mirto, Capulincillo) – understory tree
- *Psidium guajava* (Guayaba) – pioneer/edge specialist
- *Byrsonima crassifolia* (Nance; Malpighiaceae) – included for bimodal fruiting comparison

B. Lauraceae Complex:

- *Persea schiedeana* (Aguacate de montaña, Yas) – large-seeded, mammal-dispersed
- *Nectandra* spp. (Aguacatillo) – avian-dispersed, shade-tolerant

C. Structural Dominant:

- *Hauya elegans* (Guayabón, Supara; Onagraceae) – canopy emergent, hypothesized thermal facilitator

Rationale: These taxa represent a gradient of shade tolerance, seed size, and dispersal syndromes, enabling assessment of differential responses to microclimatic buffering across functional guilds.

III. MECHANISTIC MODEL

3.1 Operational Hypothesis: Four-Stage Feedback Loop

We propose a **positive ecological feedback model** comprising four interdependent stages:

- **Stage 1: Structural Infrastructure (Biogenic Cooling Architecture)**

Mechanism: Mature *Hauya elegans* canopy creates aerodynamic roughness, reducing wind velocity (predicted 40–60% reduction at 1.5 m height) and increasing boundary layer thickness. This dampens turbulent mixing and retains cooler, moister air masses at understory levels.

Testable Prediction: Wind speed at 1.5 m height will be significantly lower ($p < 0.05$) within *Hauya*-dominated nuclei compared to degraded reference areas.

- **Stage 2: Thermal Shield & Genetic Viability (Metabolic Protection)**

Mechanism: Elevated temperatures in degraded areas (>25°C) accelerate seed desiccation rates and shorten viable dormancy periods, particularly for recalcitrant seeds (e.g., *Persea*, *Nectandra*). Buffered microclimates (17–19°C) maintain metabolic quiescence, extending germination windows and reducing pre-dispersal mortality.

Testable Predictions:

1. Seed viability (germination %) will be higher for seeds collected from buffered vs. exposed microsites.
2. Germination lag time will be shorter for seeds stored under buffered conditions.
3. Early seedling survival (0–6 months) will correlate negatively with mean microsite temperature.

- **Stage 3: Phenological Investment (Resource Allocation Shift)**

Mechanism: Reduced thermal stress allows Myrtaceae and Lauraceae to allocate greater photosynthetic carbon to reproductive structures rather than stress-response proteins. This manifests as:

- Increased fruit biomass per individual
- Enhanced nutritional quality (lipid/sugar content)
- Extended fruiting windows (reduced temporal clustering)

Testable Predictions:

1. Fruit crop size (fruits per tree) will be higher in buffered sectors.
2. Individual fruit mass will be greater in buffered microsites.

3. Fruiting phenology will show lower interannual variance (higher temporal predictability) in buffered nuclei.

- **Stage 4: Avifaunal Feedback Loop (Genomic Enrichment)**

Mechanism: Consistent fruit availability attracts higher densities of avian frugivores, increasing seed deposition rates and introducing genetic material from external populations (gene flow). This enhances local genetic diversity and recruitment of heterospecific taxa not present in the baseline inventory.

Testable Predictions:

1. Bird visitation frequency (visits per hour) will correlate positively with fruit crop size in buffered nuclei.
2. Seed rain diversity (species richness) will be higher in buffered microsites.
3. Recruitment of non-baseline species will occur preferentially near *Hauya* canopy nuclei.

3.2 Integrated Model: The Genomic Sovereignty Loop

The four stages form a self-reinforcing cycle:

Structural Complexity
(H. elegans canopy)
↓
Thermal Buffering
(-10°C differential)
↓
Phenological Stability
(extended fruiting)
↓
Trophic Magnetism
(disperser attraction)
↓
Genomic Enrichment
(seed rain diversity)
↓
Enhanced Recruitment
(structural reinforcement)
↓
[CYCLE REINFORCES]

This loop transforms a forest fragment from a **demographic sink** (edge-dominated, low recruitment) into a **climatic refugium** (buffered microclimate, sustained regeneration).

IV. METHODOLOGICAL FRAMEWORK (2026–2030)

4.1 Study Design Overview

We propose a **five-year monitoring program** integrating field phenology, distributed sensor networks, camera-trap avifaunal documentation, and remote sensing validation. The study employs a **spatially stratified comparative design** with three monitoring sectors:

1. **Degraded Reference Sector** (former tajo; ~1,800 m²)
2. **Transition Zone** (regeneration buffer; ~1,000 m²)
3. **Buffered Core** (*Hauya*-dominated nucleus; ~900 m²)

All focal taxa will be monitored across sectors to assess differential responses to the thermal gradient.

4.2 Component A: Phenological Transects

Objective: Quantify fruiting intensity, fruit biomass, and phenological constancy across the thermal gradient.

Protocol:

1. **Individual Selection:** Mark ≥10 reproductive individuals per focal species using aluminum tags. Record GPS coordinates (±3 m accuracy) and initial DBH.
2. **Monthly Monitoring:** Score fruiting intensity using a semi-quantitative scale:
   - 0 = No fruit
   - 1 = 1–25% crown bearing fruit
   - 2 = 26–50%
   - 3 = 51–75%
   - 4 = 76–100%
3. **Fruit Collection:** Harvest 10–20 mature fruits per individual per fruiting episode. Measure:
   - Fresh mass (g)
   - Dry mass (48 hrs at 60°C)
   - Seed count
4. **Phenological Metrics:**
   - Fruiting duration (days per event)
   - Inter-fruiting interval (days between events)
   - Temporal predictability (coefficient of variation)

Statistical Analysis: Mixed-effects models with sector (degraded, transition, buffered) as fixed effect and individual as random effect. Response variables: fruiting intensity, fruit mass, phenological variance.

4.3 Component B: Microclimatic Gradient Monitoring

Objective: Quantify spatial and temporal variation in air temperature, soil temperature, relative humidity, and soil moisture across the degraded-to-buffered gradient.

Infrastructure:

- **Three permanent monitoring stations:**

    1. Degraded terrace (open canopy, full sun exposure)
    2. Transition zone (partial canopy, ~50% cover)
    3. Buffered core (closed canopy, *Hauya* dominance)

- **Sensor Suite per Station:**

    - Air temperature/RH datalogger (HOBO MX2301A or equivalent; 1.5 m height)
    - Soil temperature probe (10 cm depth)
    - Soil moisture sensor (10 cm depth; optional if budget permits)

Data Collection:

- Continuous logging at 30-minute intervals
- Manual download every 3 months
- Calibration checks every 6 months against reference thermometer

Derived Metrics:

- Mean daily temperature per sector
- Diurnal temperature range (DTR)
- Thermal buffering capacity: $\Delta T = T(\text{degraded}) - T(\text{buffered})$
- Interannual temperature variance (σ^2)

Statistical Analysis: Repeated-measures ANOVA comparing thermal regimes across sectors. Time-series analysis to assess seasonal stability.

4.4 Component C: Avifaunal Documentation

Objective: Quantify bird visitation rates to fruiting trees and assess correlation with fruit availability and microclimate.

Protocol:

1. **Camera Trap Deployment:** Install motion-activated cameras (Bushnell Trophy Cam or equivalent) focused on ≥5 marked fruiting individuals per focal species. Cameras positioned 3–5 m from focal tree, angled to capture crown activity.

2. **Observation Schedule:**
    - Continuous monitoring during fruiting episodes
    - 4-week deployment periods per focal tree
    - Cameras rotated among individuals to maximize spatial coverage

3. **Image Processing:**
    - Species identification via iNaturalist community verification
    - Visitation frequency (visits per hour)
    - Foraging duration (seconds per visit)

4. **Point Counts:** Supplemental dawn surveys (06:00–08:00) at each sector, conducted monthly. Record all avian species within 25 m radius.

Derived Metrics:

- Visitation rate per fruit biomass unit (visits/hr/kg fruit)
- Frugivore species richness per sector
- Temporal overlap between fruit availability and bird activity

Statistical Analysis: Generalized linear models (Poisson or negative binomial) with visitation rate as response variable; fruit biomass, sector, and temperature as predictors.

4.5 Component D: Seed Viability and Germination Trials

Objective: Test the hypothesis that thermal buffering enhances seed viability and germination success.

Protocol:

1. **Seed Collection:** Harvest ripe fruits from marked individuals in buffered and degraded sectors. Extract seeds immediately.
2. **Viability Assessment:**
    - **Tetrazolium Test:** Subsample of 50 seeds per collection batch stained with 2,3,5-triphenyltetrazolium chloride. Score viability as % viable (red-stained embryos).
3. **Germination Trials:**
    - **Common Garden Experiment:** Plant 100 seeds per species × sector combination in standardized potting mix within shadehouse at Huaravito Field Lodge.
    - Monitor germination daily for 90 days.
    - Record germination % and mean time to germination (MTG).
4. **Field Recruitment Monitoring:**
    - Establish 10 × 10 m seed rain plots (n = 5 per sector) with seed traps.
    - Census germinants and seedlings (< 50 cm height) annually.

Statistical Analysis: Chi-square tests for germination success; Cox proportional hazards models for time-to-germination; ANOVA for seedling density across sectors.

4.6 Component E: Remote Sensing Correlation

Objective: Validate ground-based observations with satellite-derived vegetation indices and assess landscape-scale patterns.

Data Sources:

- **Landsat 8/9 OLI:** 30 m spatial resolution, 16-day revisit
- **Sentinel-2 MSI:** 10 m spatial resolution, 5-day revisit
- **MODIS:** Terra/Aqua products for regional climate context

Derived Indices:

- **NDVI (Normalized Difference Vegetation Index):** Proxy for photosynthetic activity and canopy greenness
- **EVI (Enhanced Vegetation Index):** Less sensitive to atmospheric effects
- **LST (Land Surface Temperature):** MODIS thermal bands for regional temperature context

Analysis:

1. Extract mean NDVI/EVI for each monitoring sector (degraded, transition, buffered) across the 5-year time series.
2. Test hypothesis: Buffered nuclei maintain higher NDVI during regional dry-season stress events (indicated by regional LST anomalies).
3. Correlate ground-based temperature data with MODIS LST to validate microclimatic decoupling.

Expected Outcome: Buffered nuclei will exhibit delayed senescence (sustained greenness) during regional drought events, corroborating ground-based thermal stability observations.

V. EXPECTED OUTCOMES AND SIGNIFICANCE

5.1 Scientific Contributions

This research will provide the first mechanistic assessment of how **structural canopy dominants** (specifically *Hauya elegans*) mediate microclimatic buffering and phenological stability in Neotropical premontane forests. Key contributions include:

1. **Quantification of thermal buffering magnitude** across a degraded-to-intact gradient, enabling comparative analysis with other tropical forest systems.
2. **Mechanistic linkage** between forest structure, microclimate, and reproductive phenology, addressing a critical knowledge gap in tropical restoration ecology.
3. **Functional metric for climatic refugium status:** Interannual temperature variance (σ) as a low-cost, high-signal indicator of ecosystem resilience.
4. **Baseline dataset** for long-term monitoring of climate change impacts on premontane forest phenology.

5.2 Alignment with SNAPP-MORE Objectives

This research directly addresses SNAPP-MORE's core objectives:

- **Early monitoring indicators:** Thermal stability (σ ~0.2°C) as a predictor of long-term restoration trajectory.
- **Cost-effectiveness:** Distributed sensor networks and citizen science platforms (iNaturalist, GLOBE Observer) minimize field costs while maintaining data quality.
- **Scalability:** Methodology is replicable across fragmented tropical landscapes with minimal infrastructure requirements.
- **Policy relevance:** Findings will inform identification and prioritization of climate refugia for conservation under regional warming scenarios.

5.3 Conservation Implications

If the Phenothermic Synergy Hypothesis is supported, *Hauya elegans*-dominated forest patches may represent **high-priority conservation targets** as climatic refugia. Management implications include:

- **Prioritization of structurally intact forest nuclei** in landscape-scale conservation planning.
- **Retention of large-diameter canopy trees** (especially *H. elegans*) in restoration projects to maximize microclimatic buffering.
- **Assisted colonization** of *H. elegans* into degraded sites to accelerate thermal stabilization and phenological recovery.

VI. STRATEGIC POSITIONING: THE LTEMS FIELD LODGE

6.1 Infrastructure Integration

The **Huaravito Reserve LTEMS Field Lodge** serves dual functions:

1. **Observation Platform:** The Lodge is positioned at the degraded-buffered ecotone, providing direct visual and sensor access to the thermal gradient.
2. **Experimental Control:** The degraded terrace (former clay extraction zone) functions as the comparative baseline for all monitoring activities.

This configuration transforms the Field Lodge from passive accommodation into an **active component of the experimental design**.

6.2 Operational Advantages

- **Proximity to monitoring sites:** All three sectors (degraded, transition, buffered) accessible within 150 m of Lodge.
- **Power and connectivity:** Solar panels and internet access enable continuous datalogger operation and real-time data upload.
- **Visitor engagement:** Monitoring infrastructure visible to Lodge guests, facilitating science communication and potential collaborations.

VII. TIMELINE AND DELIVERABLES (2026–2030)

Year 1 (2026):

- Infrastructure establishment (sensor deployment, tree tagging)
- Baseline phenological census
- Initial seed viability trials
- Quarterly progress reports to SNAPP-MORE

Year 2 (2027):

- Full phenological monitoring (all focal species)
- Camera trap avifaunal documentation
- First-year germination trial results
- Interim data analysis and manuscript preparation

Year 3 (2028):

- Re-measurement of permanent plots
- Second seed viability cohort
- Remote sensing time-series analysis
- Mid-term synthesis report

Year 4 (2029):

- Continued monitoring
- Statistical modeling and hypothesis testing
- Draft manuscripts for peer-reviewed journals
- Stakeholder workshops (UCR, SINAC, OTS)

Year 5 (2030):

- Final data synthesis
- Completion of 5-year Technical Synthesis Report
- Publication of 2–3 peer-reviewed articles
- Development of monitoring protocol guide for replication

VIII. BUDGET ESTIMATE (PRELIMINARY)

1 Equipment & Sensors

Item	Units	Cost (USD)	Justification
Temperature / RH dataloggers	3	900	Continuous microclimate monitoring.
Soil probes	3	450	Soil moisture & temperature near focal species.
Camera traps	5	1,350	Document avian dispersers & interactions.
Diameter tape, clinometer, GPS	1 set	300	Tree measurements and spatial mapping.
Subtotal		**3,000**	

2 Field Operations

Item	Units / Duration	Cost (USD)	Justification
Monthly monitoring visits	60 visits × $50	3,000	On-site phenology & seed trials.
Seed germination trials	Supplies	500	Baseline germination & microclimatic effects.
Subtotal		**3,500**	

3 Data Management & Analysis

Item	Units	Cost (USD)	Justification
Remote sensing software licenses	1	500	NDVI & canopy correlation with microclimate.
Statistical consulting	1	1,000	Advanced analysis of phenology, thermal differentials & disperser interactions.
Subtotal		**1,500**	

4 Personal Compensation (Food Minimum Living Cost)

Item	Duration	Cost (USD)	Justification
Principal investigator subsistence	60 months × $200/month	12,000	Covers realistic food basic living expenses in Costa Rica for 5 years.

TOTAL 5-YEAR BUDGET: $20,000 USD

IX. INSTITUTIONAL PARTNERSHIPS

We seek collaborative partnerships with:

- **Universidad de Costa Rica (UCR)** – Escuela de Biología
- **Universidad Nacional (UNA)** – Programa de Ecología y Recursos Naturales
- **Organization for Tropical Studies (OTS)** – Fieldwork logistics and academic oversight
- **CATIE (Centro Agronómico Tropical)** – Agroforestry and restoration expertise
- **EARTH University** – Undergraduate research opportunities

These partnerships would provide:

- Taxonomic verification and voucher specimen deposition
- Statistical and analytical support
- Access to laboratory facilities for seed viability assays
- Co-authorship on publications

X. DATA MANAGEMENT AND SHARING

All data will be managed according to FAIR principles (Findable, Accessible, Interoperable, Reusable):

- **Primary Data Repository:** Huaravito Reserve LTEMS archive (CSV format with metadata)
- **Public Platforms:**
  - iNaturalist (species observations)
  - Terraware (restoration monitoring)
  - NASA GLOBE Observer (biometric data)
  - GBIF (biodiversity occurrence records)
- **Data Sharing Agreement with SNAPP-MORE:**
  - Raw datasets provided upon request
  - Anonymization of site-specific GPS coordinates in published datasets (regional-level precision only)
  - Opportunity for Custodian-Director to review manuscripts prior to submission

XI. RISKS AND LIMITATIONS

11.1 Methodological Limitations

- **Spatial Pseudoreplication:** Single site; findings may not generalize to other *Hauya elegans* populations or premontane forests.
    - **Mitigation:** Comparative analysis with published microclimatic datasets; emphasis on mechanistic understanding over site-specific conclusions.
- **Observer Bias:** Single-observer phenological monitoring may introduce systematic error.
    - **Mitigation:** Photographic vouchers for all phenological events; periodic training with external botanists.
- **Climate Variability:** ENSO events or extreme weather may obscure treatment effects.
    - **Mitigation:** 5-year time series provides sufficient temporal replication to detect trends despite inter-annual noise.

11.2 Operational Risks

- **Equipment Failure:** Datalogger malfunction or theft could result in data gaps.
    - **Mitigation:** Redundant sensors; monthly manual temperature readings as backup; tamper-proof enclosures.
- **Funding Interruption:** Lack of sustained support could compromise continuity.
    - **Mitigation:** Phased design allows partial implementation if full budget unavailable; core monitoring (temperature, phenology) prioritized.

XII. CONCLUSION

The **Phenothermic Synergy Hypothesis** proposes that microclimatic buffering mediated by structural canopy dominants like *Hauya elegans* represents a critical but underappreciated mechanism for maintaining phenological stability and genetic viability in fragmented tropical forests. The Huaravito Reserve LTEMS provides an exceptional natural laboratory to test this hypothesis, with documented thermal gradients of sufficient magnitude to enable robust comparative analysis.

This research directly addresses SNAPP-MORE's mandate to identify cost-effective, early-stage monitoring indicators that predict restoration success. We propose that **thermal stability (interannual variance σ ~0.2°C)** functions as such an indicator, signaling the presence of intact ecological feedback loops that sustain trophic interactions and genomic flow despite external climatic stress.

Beyond scientific merit, this research positions the Huaravito Reserve LTEMS as a model for **community-governed ecological observatories**, demonstrating that rigorous long-term monitoring can be sustained through local stewardship integrated with global data platforms (NASA GLOBE, iNaturalist, Terraware). By bridging local knowledge systems (Huaravito Reserve – Sacred Natural Site and Toyopán Sacred Forest designation) with aerospace technology (satellite validation), the project exemplifies biocultural approaches to conservation science in the Anthropocene.

We invite you to consider this proposal as a pilot study for a broader network of thermally-stratified monitoring sites across the Neotropical premontane zone, with potential expansion to other climate-sensitive ecosystems globally.

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